



Chapter 8 Case Study

Data Analytics Using Python

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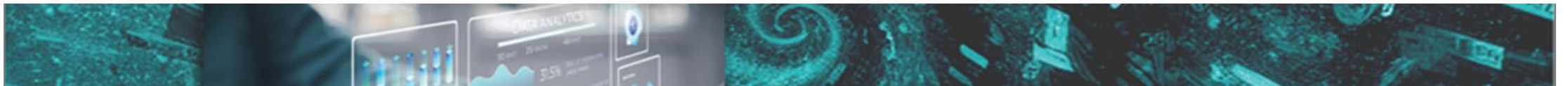
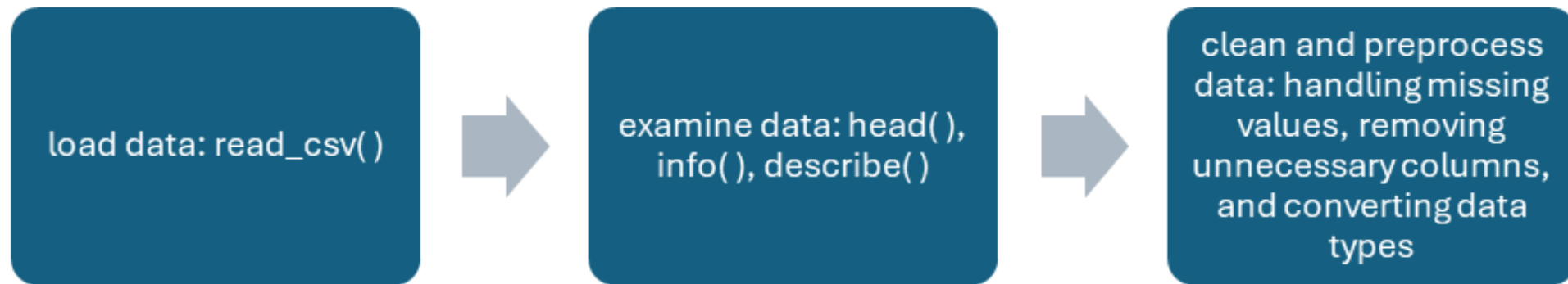


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<https://he.kendallhunt.com/product/data-analytics-using-python>

Three stages of Data Analysis

- first stage: load the raw data into a dataframe, examine the dataframe to understand the dataset, and preprocess and clean the dataset



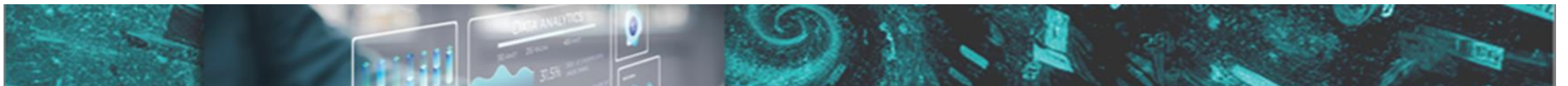
Three stages of Data Analysis

- second stage: conduct exploratory data analysis, which involves identifying trends, patterns, or anomalies in the distribution and relationships of data points

data wrangling

data aggregation

visualization:
correlation heatmap,
bar plot, scatter plot,
line plot, etc.



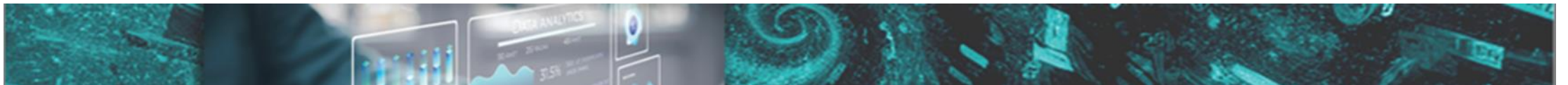
Three stages of Data Analysis

- third stage: data modeling
 - use either statistical models or machine learning models to quantify relationships among variables, identify correlations, and assess model performance

linear regression
model for
relationship on
continuous data

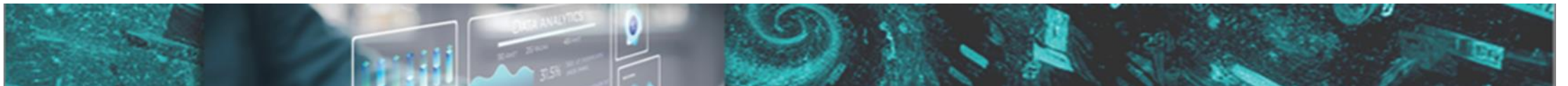
logistic regression
model for data
classification

clustering for
separating data
into groups



Kaggle Company Income Analysis

- contains quarterly income statements for over 3,600 stocks listed on the NYSE and NASDAQ, each with a market capitalization greater than 100 million USD. With nearly 70,000 entries and more than 40 columns, this dataset presents a rich resource for exploration and analysis of interesting patterns among listed companies



Sleep Health and Lifestyle Analysis

- provides comprehensive data on sleep health and lifestyle factors, including gender, age, occupation, sleep duration, quality of sleep, physical activity level, stress levels, BMI category, blood pressure, heart rate, daily steps, and the presence or absence of sleep disorders. The dataset contains 400 entries and 13 columns.

